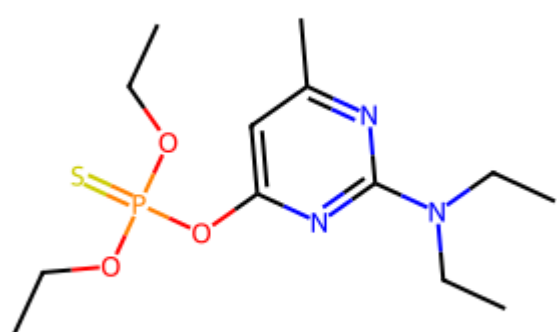
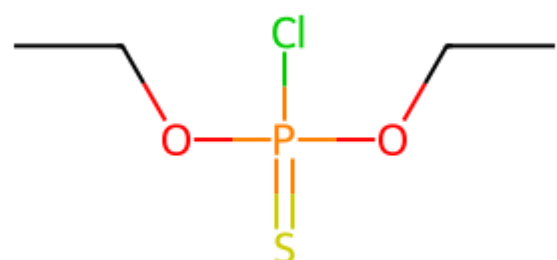


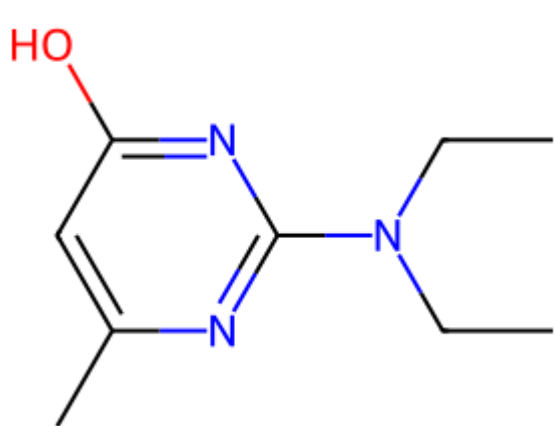


{'analysis_time': 0.0,
'first_solution_time': 1.214405,
'num_calls_rxn_model': 0.0,
'num_calls_value_function': 0.0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 5.943280842856512,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 5.943280842856512,
'retro_star_value': 5.943280842856512}

Reaction
{'analysis_time': 0.196977,
'first_solution_time': 1.214405,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 5.943280842856512,
'retro_star_reaction_number': 5.943280842856512,
'retro_star_rxn_cost': 0.03244708539974736,
'retro_star_value': 5.943280842856512}

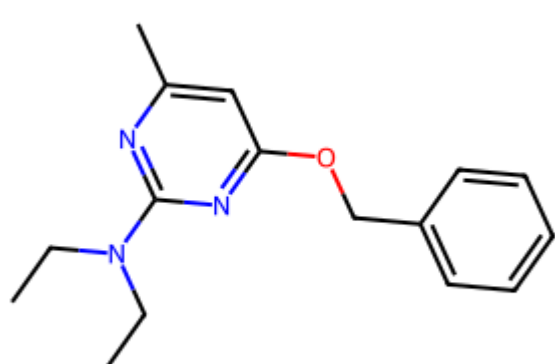


{'analysis_time': 0.196977,
'first_solution_time': 0.196977,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 5.943280842856512}



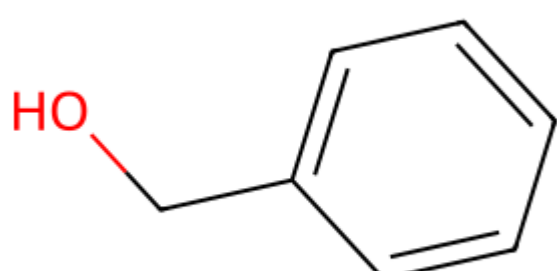
{'analysis_time': 0.196977,
'first_solution_time': 1.214405,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0.0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 5.910833757456765,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 5.910833757456765,
'retro_star_value': 5.943280842856512}

Reaction
{'analysis_time': 0.332676,
'first_solution_time': 1.214405,
'num_calls_rxn_model': 2,
'num_calls_value_function': 37,
'retro_star_min_cost': 5.910833757456765,
'retro_star_reaction_number': 5.910833757456765,
'retro_star_rxn_cost': 2.3366279198703894,
'retro_star_value': 5.943280842856512}

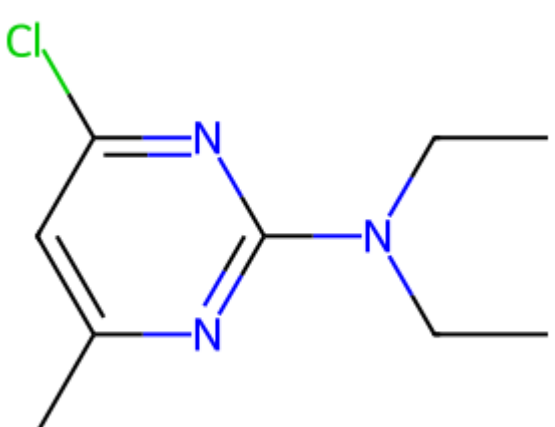


{'analysis_time': 0.332676,
'first_solution_time': 1.214405,
'num_calls_rxn_model': 2,
'num_calls_value_function': 37,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 3.5802058375863757,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 3.5802058375863757,
'retro_star_value': 5.943280842856512}

Reaction
{'analysis_time': 1.212884,
'first_solution_time': 1.214405,
'num_calls_rxn_model': 8,
'num_calls_value_function': 155,
'retro_star_min_cost': 3.5802058375863757,
'retro_star_reaction_number': 3.5802058375863757,
'retro_star_rxn_cost': 0.01005033585350145,
'retro_star_value': 5.943280842856512}

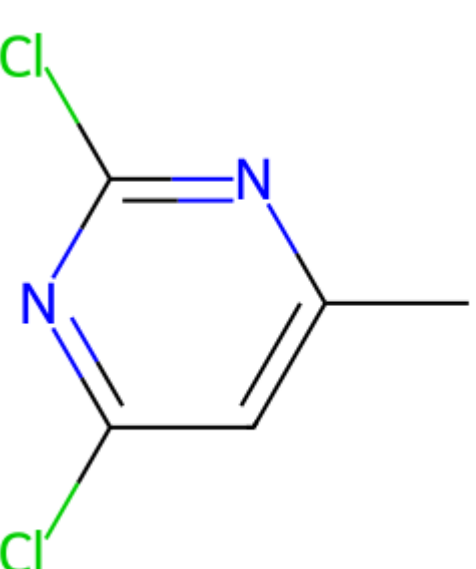


{'analysis_time': 1.212884,
'first_solution_time': 1.212884,
'num_calls_rxn_model': 8,
'num_calls_value_function': 155,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 5.943280842856512}

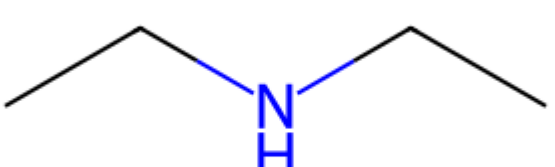


{'analysis_time': 1.212884,
'first_solution_time': 1.214405,
'num_calls_rxn_model': 8,
'num_calls_value_function': 155,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 3.570155501732874,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 3.570155501732874,
'retro_star_value': 5.943280842856512}

Reaction
{'analysis_time': 1.214405,
'first_solution_time': 1.21561,
'num_calls_rxn_model': 8,
'num_calls_value_function': 179,
'retro_star_min_cost': 3.570155501732874,
'retro_star_reaction_number': 3.570155501732874,
'retro_star_rxn_cost': 0.01005033585350145,
'retro_star_value': 5.943280842856512}

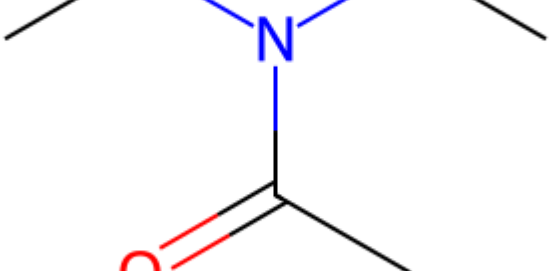


{'analysis_time': 1.214405,
'first_solution_time': 1.214405,
'num_calls_rxn_model': 8,
'num_calls_value_function': 179,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 5.943280842856512}



{'analysis_time': 1.214405,
'first_solution_time': 1.21561,
'num_calls_rxn_model': 8,
'num_calls_value_function': 179,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 3.5601051658793725,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 3.5601051658793725,
'retro_star_value': 5.943280842856512}

Reaction
{'analysis_time': 1.21561,
'first_solution_time': 1.21561,
'num_calls_rxn_model': 8,
'num_calls_value_function': 200,
'retro_star_min_cost': 3.5601051658793725,
'retro_star_reaction_number': 3.5601051658793725,
'retro_star_rxn_cost': 3.5601051658793725,
'retro_star_value': 5.943280842856512}



{'analysis_time': 1.21561,
'first_solution_time': 1.21561,
'num_calls_rxn_model': 8,
'num_calls_value_function': 200,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 5.943280842856512}